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Optimization of Low Reynolds Number Wing's Spanwise Lift Distribution



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Introduction

In continuation to the previous project of designing Low Reynolds Number UAV, this study will be focused on optimizing the wing's geometry to decrease the induced drag. Ideal lift distribution is produced by an Elliptical Wing Planform, reducing the lift induced drag via concentrating most of the lift inboard of the wing, which in turn reduces the tip vortices. While elliptical wing is hard to manufacture and have tip stall first behavior which impacts roll authority and spin recovery, it can be approximated by introducing taper to the wing.

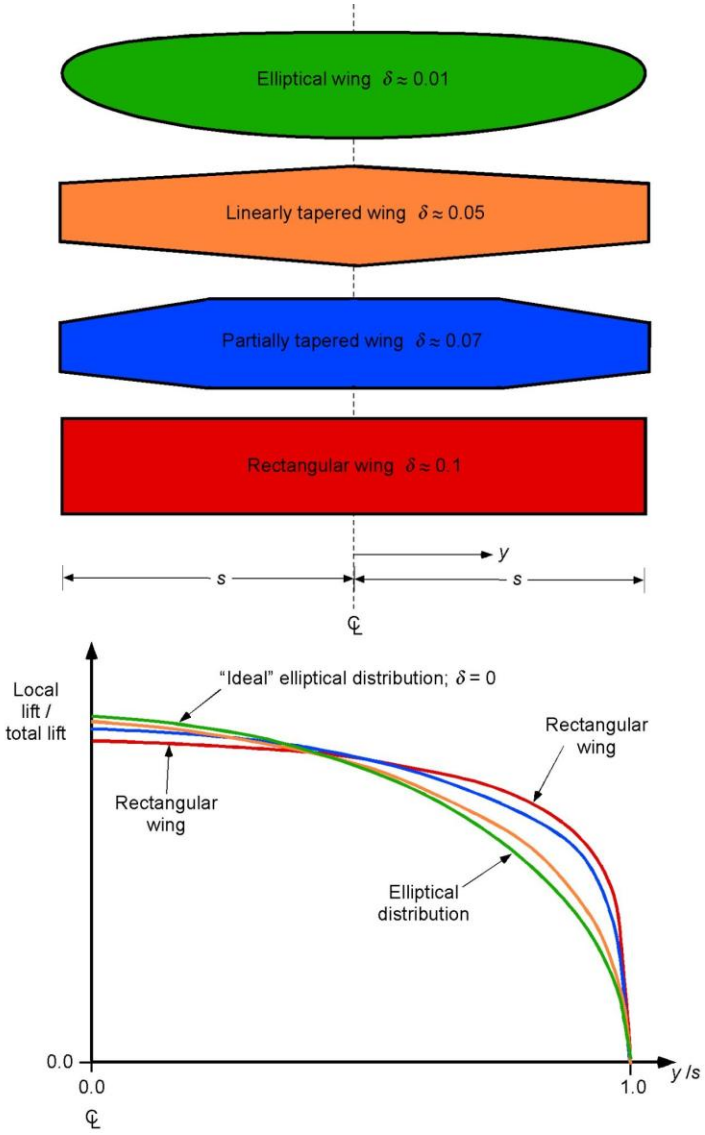


Figure 1: Lift Distribution for Various Wing Planforms

Objectives

•Simulate clean Wing-Body UAV of the current design, and analyze its lift distribution to generate the new wing planform.

•Compare the lift distribution between the old and new wing design at cruise speed.

•Calculate the reduction in drag and bending moment distribution required for structural design.

Rectangular Wing

Since mesh dependency test were already conducted in the previous project, same settings will be used for both the rectangular wing and the tapered one.

Contours and Pathlines

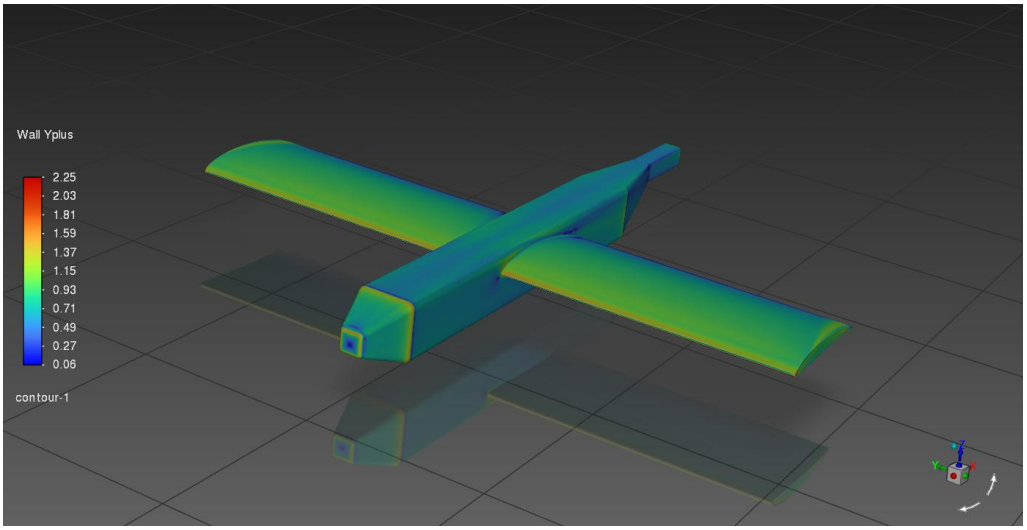


Figure 2:y+ Contour

•The y+ contour shows values lower than 5.

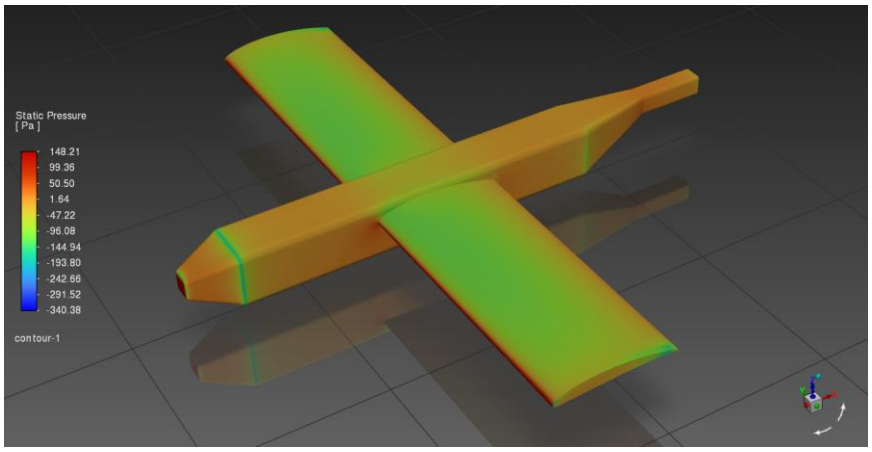


Figure 3: Pressure contour

The pressure contour shows show no significant behaviors, except highlighting the low-pressure region at the aft of the wingtip, this low-pressure zone contributes to the spike of lift and drag at the tip section, caused by interaction between the tip vortex and the trailing edge of the wingtip. Being at the aft region of the wingtip, it creates an additional force vector in the drag direction.

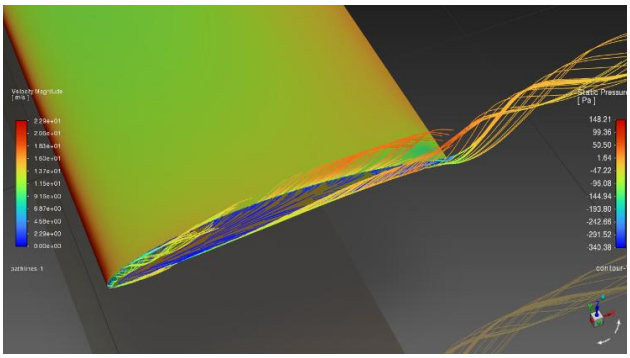


Figure 4: Tip Vortex interacting with wingtip.

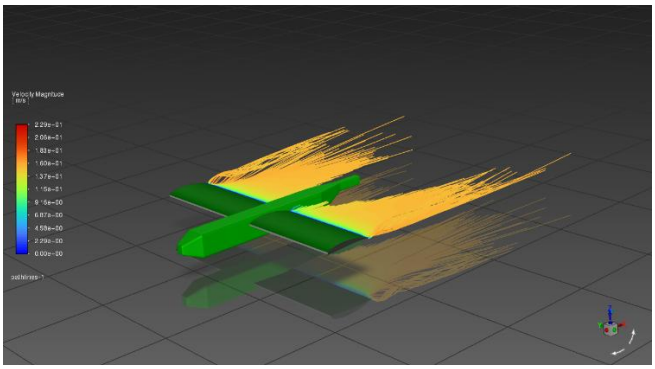


Figure 5:Wake sheet and tip vortex

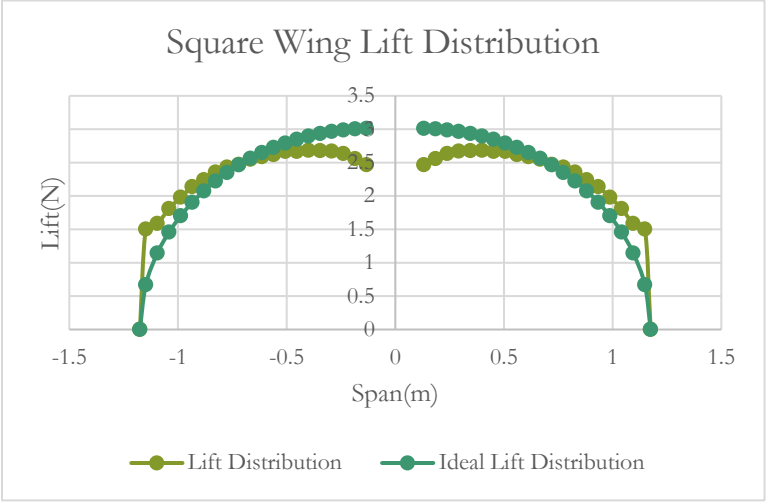


Figure 6: Lift distribution of Rectangular wing.

The Lift distribution of the rectangular wing shows that it generates less lift than the ideal one before 0.72m span. And generates more after it, with spike of lift at the tip due to the afore mentioned reasons. From this plot we can deduce the required geometry of the new wing.

New wing design calculations

The equation that governs the elliptical wing distribution is as follows:

$$c(y) = c_0 \cdot \sqrt{1 - \frac{2y}{b}}$$

Because the lift generated at 0.72m is the same for the elliptical and rectangular wing, then the chord at this location is the same. Thus, we can calculate the root chord c_o, then we can proceed with the rest of the geometry.

Parameter	Rectangular	New Design.
C_o	0.47m	0.59m
$c_{0.72}$	0.47m	0.47m
C_{tip}	0.47m	0.2m
$\lambda_{b<0.72}$	1	0.8
$\lambda_{b>0.72}$	1	0.47
$\bar{C}$	0.47m	0.485m

No Leading-edge sweep were used to prevent the loss of Reynolds number on the wing, so that the wing design speed matches the rectangular wing.

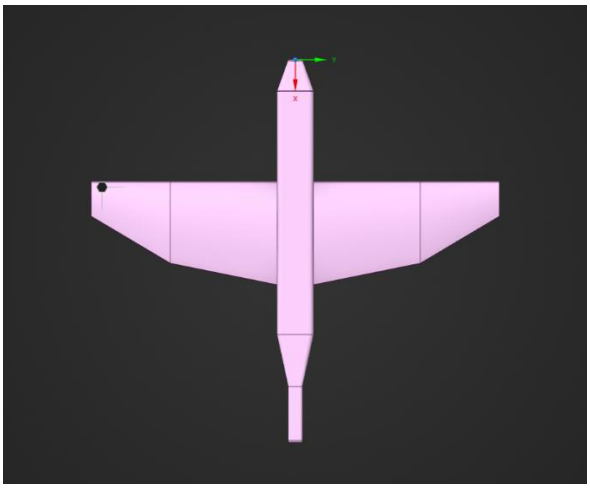


Figure 7: New Wing Geometry

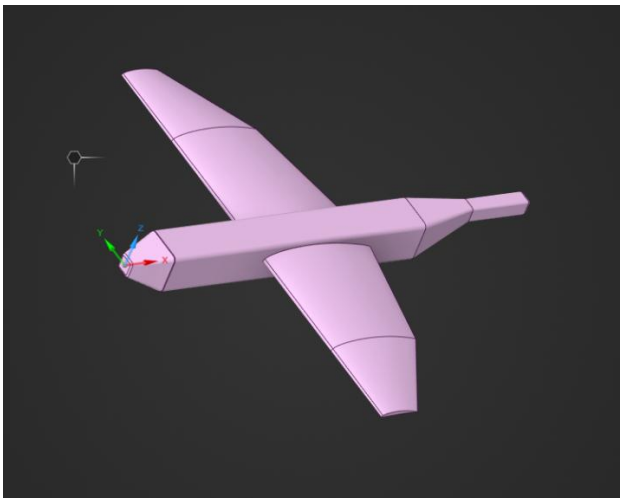


Figure 8: UAV

The new design features a double tapered wing, that closely mimics the elliptical wing, the increased chord at the root is to generate more lift at the inboard region, while the outboard chord is reduced to reduce the amount of lift generated there.

Double Tapered Design CFD

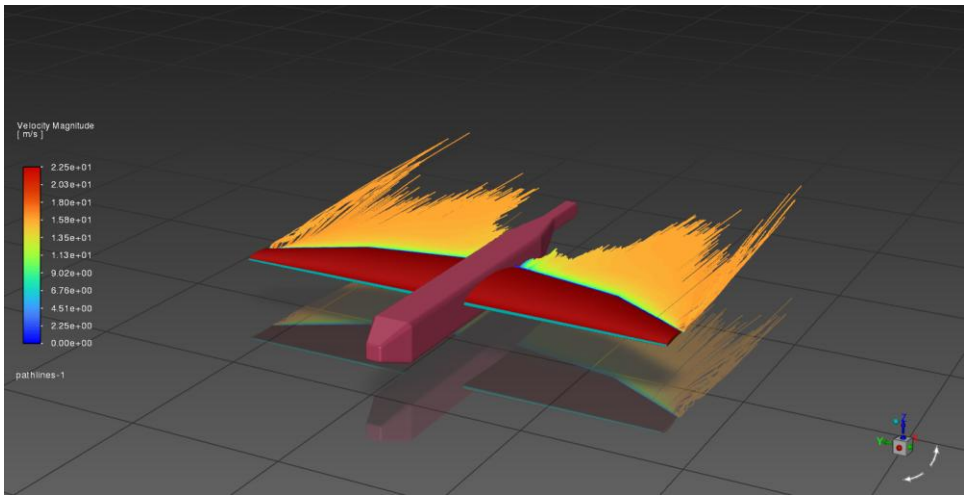


Figure 9:Pathlines

The pathlines shows a weaker tip vortex "not very noticeable unless we zoom in" and a gentler wake sheet curl up, indicating minimized energy in the wake caused by lift induced drag.

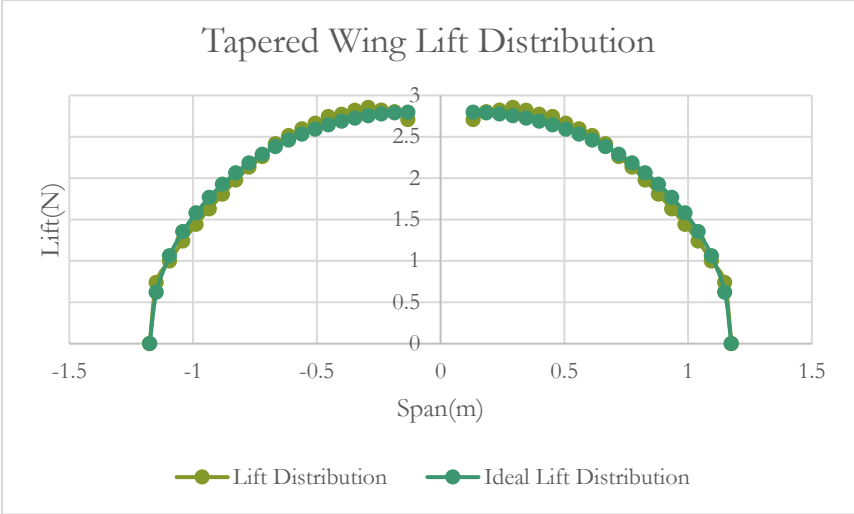


Figure 10:Tapered Wing Lift Distribution

The lift distribution of the new design almost coincides with the ideal case, indicating that the wing has reached its maximum efficiency, also we can see that the spike of lift at the tip is no longer there.

Drag and bending moment distribution.

Drag

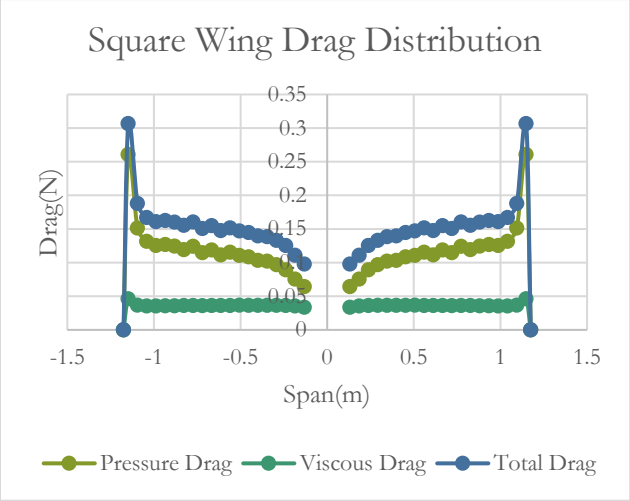


Figure 11:Drag Distribution of Rectangular Wing

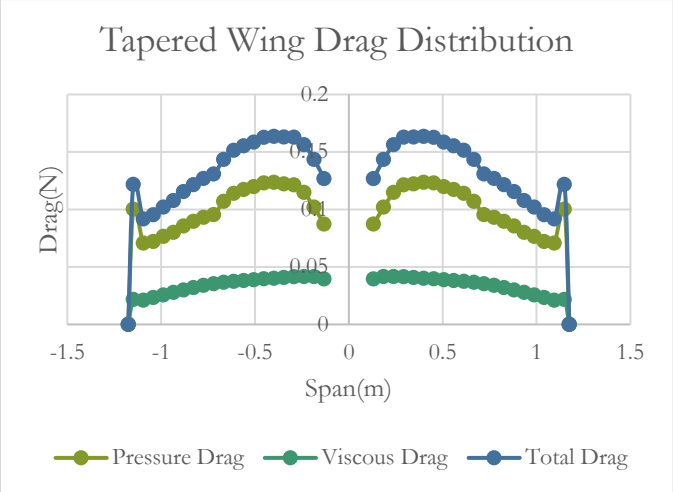


Figure 12:Drag Distribution of Tapered Wing

The drag distribution plots show that the tip induced drag decreased from 0.33N to 0.12N, that's over 50% reduction of induced drag. While also the drag in the inboard section increased slightly due to the increased profile area.

The total drag of both wings are 6.2N for the rectangular wing and 5.4N for the tapered wing. Indicating a whopping 13% drag reduction.

But to be fair the drag reduction cam at a cost, the wing now generates less lift that the rectangular wing at 15.5m/sec, to compare them side to side we need to compare them at the speed where they generate sufficient lift for the MTOW of 10Kg, this speed was 16m/sec. comparing them together showed that the drag reduction is 9%, while also flying faster now.

Bending Moment.

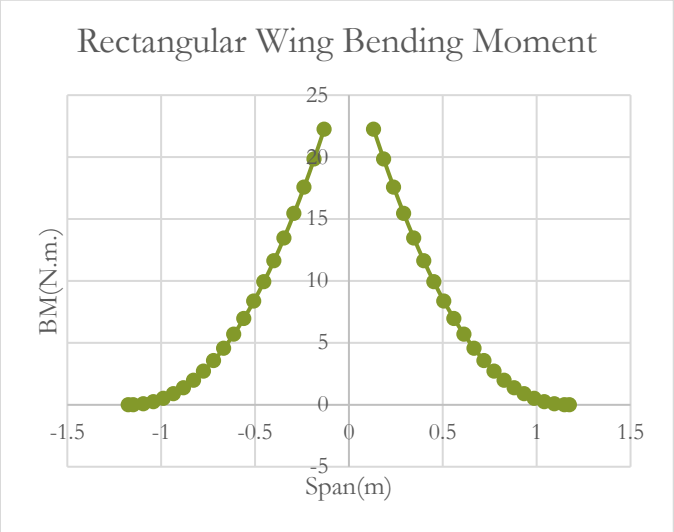


Figure 13:Rectangular Wing Bending

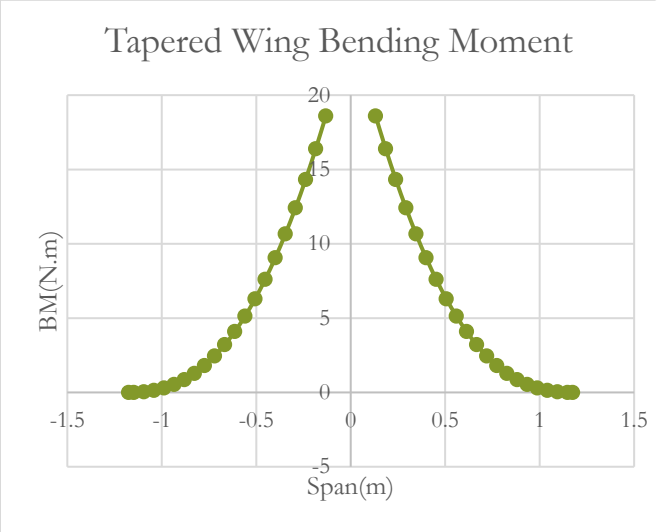


Figure 14:Tapered Wing Bending Moment

The maximum bending moment at the root was reduced from 22.25NM to 18.5NM. that's 16.85% reduction in structural stresses, that would decrease the need for a beefier spar, which will reduce structural weight in return.

Summary of results

Parameter	Rectangular wing	Tapered Wing	Benefit
Cruise Velocity	15.5 m/sec	16 m/sec	3.23%
Drag	6.2N	5.642N	9%
Bending Moment at Root	22.25NM	18.5NM	16.85%